

Name: _____ Date: _____

Period: _____

Primary Objective

By the end of this lesson, I will be able to...

1. Describe how $g(x) = af(x + h) + k$ transforms the graph of f using **translations** and **dilations (and reflections)**. (LO 1.12.A)
2. Determine the **domain** and **range** of a transformed function from the parent's domain and range. (LO 1.12.A)

Vocabulary & Key Concepts

parent function:

The original, un-transformed function from which transformed versions are built......

transformation:

An operation that shifts, scales, or reflects the graph of a function...... **vertical translation:**

$g(x) = f(x) + k$: **shifts the graph up** ($k > 0$) **or down** ($k < 0$) **by** $|k|$ **units.**... **horizontal translation:**

$g(x) = f(x + h)$: **shifts the graph left** ($h > 0$) **or right** ($h < 0$) **by** $|h|$ **units.**..... **vertical dilation:**

$g(x) = af(x)$: **stretches** ($|a| > 1$) **or compresses** ($0 < |a| < 1$) **outputs;** $a < 0$ **also reflects over** x -axis.

horizontal dilation:

$g(x) = f(bx)$: **compresses** ($|b| > 1$) **or stretches** ($0 < |b| < 1$) **inputs;** $b < 0$ **also reflects over** y -axis. ...

reflection:

A flip of the graph over an axis; $a < 0$ **flips over** x -axis; $b < 0$ **flips over** y -axis.

Hook

One function — four disguises. Can you un-transform it back?

Each transformation below produces the *same* parent shape in a different location or size. Your job today is to decode the rule so you can construct *any* disguise or reverse-engineer it from a table.

Type 1 — Vertical Translation: $g(x) = f(x) + k$

Rule (EK 1.12.A.1): Adding k to the output **shifts (translates)** the graph by k units.

- $k > 0$: shift **up** $k < 0$: shift **down**

Example: Let $f(x) = x^2$. Let $g(x) = f(x) + 3 = x^2 + 3$.

x	$f(x) = x^2$	$g(x) = f(x) + 3$
-2	4	7
-1	1	4
0	0	3
1	1	4
2	4	7

Effect on domain/range: Domain **unchanged**. Range shifts **up by 3**.

Type 2 — Horizontal Translation: $g(x) = f(x + h)$

Rule (EK 1.12.A.2): Replacing x with $x + h$ shifts the graph **horizontally** by $|h|$ units.

- $h > 0$: shift **left** $h < 0$: shift **right**

Caution: $g(x) = f(x + 3)$ shifts the graph **left 3** (not right), because we need $x + 3 = 0$, so $x = -3$.

Example: $f(x) = x^2$, $g(x) = f(x - 2) = (x - 2)^2$.

x	$f(x) = x^2$	$g(x) = f(x - 2)$
0	0	4
1	1	1
2	4	0
3	9	1
4	16	4

Effect on domain/range: Domain shifts **right by 2**. Range **unchanged**.

Type 3 — Vertical Dilation: $g(x) = af(x)$, $a \neq 0$

Rule (EK 1.12.A.3): Multiplying the *output* by a scales the graph **vertically** by $|a|$.

- $|a| > 1$: **vertical stretch** (stretches) $0 < |a| < 1$: **vertical compression** (compresses)
- $a < 0$: also reflects over the **x -axis**

Example: $f(x) = x^2$; $g(x) = 2f(x) = 2x^2$; $h(x) = -f(x) = -x^2$.

x	$f(x) = x^2$	$g(x) = 2f(x)$	$h(x) = -f(x)$
-2	4	8	-4
-1	1	2	-1
0	0	0	0
1	1	2	-1
2	4	8	-4

Effect on domain/range: Domain **unchanged**. Range is multiplied by a .

Type 4 — Horizontal Dilation: $g(x) = f(bx)$, $b \neq 0$

Rule (EK 1.12.A.4): Multiplying the *input* by b scales the graph **horizontally** by $\frac{1}{|b|}$.

- $|b| > 1$: **horizontal compression** (compresses horizontally) $0 < |b| < 1$: **horizontal stretch** (stretches horizontally)
- $b < 0$: also reflects over the **y -axis**

Example: $f(x) = x^2$; $g(x) = f(2x) = (2x)^2 = 4x^2$.

x	$f(x) = x^2$	$g(x) = f(2x)$
-1	1	4
$-\frac{1}{2}$	$\frac{1}{4}$	1
0	0	0
$\frac{1}{2}$	$\frac{1}{4}$	1
1	1	4

Combined Transformations: $g(x) = af(x+h)+k$

Rule (EK 1.12.A.5): Apply transformations in this order:

Step 1: Horizontal dilation/reflection (b) **Step 2:** Horizontal shift (h)

Step 3: Vertical dilation/reflection (a) **Step 4:** Vertical shift (k)

Example: Describe all transformations applied to f to produce $g(x) = -2f(x+1) - 3$.

Parameter	Transformation
$a = -2$	Vertical stretch by factor 2; reflect over x-axis
$h = 1$	Horizontal shift left 1 unit
$k = -3$	Vertical shift down 3 units

Effect on domain/range (EK 1.12.A.6): If the domain of f is $[-2, 4]$ and the range of f is $[0, 9]$, then:

Domain of g : $[-3, 3]$ Range of g : $[-21, -3]$

Teacher Note

Domain shifts left 1: $[-2-1, 4-1] = [-3, 3]$.

Range: multiply by -2 gives $[-18, 0]$, then shift down 3 gives $[-21, -3]$. Note the interval flips because multiplying by $-2 < 0$ reverses the inequality.